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Kinetics of Melting in Microgravity
The kinetics of melting pivalic acid (PVA) dendrites was observed under convection-free conditions on STS-87 as part of the United States Microgravity Payload Mission (USMP-4) flown on Columbia in 1997. Analysis of video data show that PVA dendrites melt without relative motion with respect to the quiescent melt phase. Dendritic fragments display shrinking to extinction, with fragmentation occurring at higher initial supercoolings. The microgravity melting kinetics against which the experimental observations are compared is based on conduction-limited quasi-static melting under shape-preserving conditions. Agreement between analytic theory and our experiments is found when the melting process occurs under shape-preserving conditions as measured using the C/A ratio of individual needle-like crystal fragments.

Video data from USMP-4 have been analyzed to establish the rate of melting of individual crystallites in the dendritic

Task Description:

mushy zone. The analysis uses potential theoretic methods to calculate analytic descriptions of the thermal field surrounding a prolate spheroidal crystal of PVA subject to melting. The new analysis (Glicksman et al., 2003) is the first that describes quasi-static melting under convection-free conditions. Direct comparison between the video data and the analytical predictions shows excellent correspondence. The IDGE data provided a unique opportunity to observe melting of dendrites without the complications of sedimentation or buoyancy-induced convection.

Mushy Zones Evolution

We analyzed two PVA crystals melting in microgravity, each formed originally as dendrite created in separate IDGE growth cycles. These dendrites grew from pure molten pivalic acid that was supercooled. The mushy zones consisted of approximately 4\% solid when melting was initiated. The time needed to melt this mushy zone completely was approximately 12 minutes. The last large fragments to melt were selected for quantitative analysis. These fragments had initial lengths of C0=0.760cm and 0.695cm, respectively.

To calculate the melting kinetics of these fragments one requires the C/A ratio, the thermal diffusivity of the melt, and the Stefan number imposed by the melting cycle. The thermistors arranged within the IDGE thermostat were optimized for measuring the supercooling prior to dendritic growth quite precisely (±0.002K). The subsequent temperature history within the growth chamber, especially during the melting portion of an experimental growth cycle, is, unfortunately, less certain, because temperature uncertainties arose from the lack of synchronization between the video clock and the mission clock. In view of this situation, one may instead adopt a self-consistent method of estimating the Stefan number in the melt during each melting sequence. The basis of the value of the self-consistent Stefan number is found from the C/A ratio (which equaled 12 for the entire melt cycle), the inital length, C0=0.760, and the length C(t) at a later time t where the fragment has decreased due to melting. In this instance the process of melting was followed to extinction, so (t)=0, and t*=40s.

The crystal melting data were fit in their entirety using the self-consistent Stefan number, which corresponds to the melt being superheated 0.63K above its equilibrium melting point, Tm=35.97K. The melting kinetics predicted over the entire the lifetime of this dendrite fragment using potential theory approach is in good agreement with experiment. The IDGE video data show only small disparities from the theoretical predictions. The second crystal analyzed, initially formed at a supercooling of 0.421K, did not melt with a constant C/A ratio. These melting data were “sectorized” into 5 sequential melting periods of 10 seconds each, except for the last sector, which was 4 seconds long. Sectorization allowed calculation of the C/A ratio, which was changing in this case. The same process of selecting a Stefan number was used, except each sector of melting was independently calculated using the initial and final lengths for that sector.

Tip–shape analysis for PVA

The analyses of USMP-4 data conducted to date indicate that the regression techniques, already used successfully to extract radius of curvature and shape of the tip of SCN dendrites, fail for pivalic acid dendrite tips recorded on USMP-4. In short, it has been found that 4th-order polynomials do not properly describe PVA tip shapes, and cannot provide consistent and reliable radii of curvature at the tip. The robustness of such methods is then judged by the invariance achieved in the radii values computed, versus the size of the interfacial sampling region in the measurement. Current efforts to obtain radii data to characterize a dendrite tip now use an older, one-parameter method of fitting the dendritic profile over various interface ranges to a perfect parabola. The canonical value for R is found by extrapolating the derived shape information back to the tip itself.

Although all dendrites share strong similarities, and possess tips that appear to be parabolic—at least to first order—there nevertheless seems to be numerous ways to characterize them. At present, it is certainly far from clear as to how many parameters are required to describe the tip shape of PVA dendrites. This shortcoming remains where we desire to characterize a dendrite over its entire smooth, branchless region, rather than at the exact tip itself. It is clear that robust methods to extract crucial tip–radii values and tip–shape characteristics are both much needed yet lacking for anisotropic materials like pivalic acid. PVA continues to be one of the most experimentally well studied dendrite–forming systems, and one of only two systems where pure diffusion data is available from microgravity. Thus, it seems important to explore this issue more fully and characterize properly the interfacial shape of PVA dendrites grown under diffusion control.

IDGE video data

Perhaps the most exciting hardware modification to the IDGE was the addition of video storage for USMP-4, capable of on–board recording dendritic growth as 256–level grayscale video, 640x480 pixels, at 30 electronic frames per second. This hardware modification was planned primarily for demonstration purposes and technology advancement. None of the so–called “primary IDGE science” involved the acquisition of video data, although much was learned, particularly about the dynamics of dendritic growth and the kinetics of conduction-limited melting of mushy zones.

RIDGE scientists have examined PVA video data and searched for subtle dynamic features such as the presence of dominant eigenfrequencies appearing in the dendrite growth velocity—a phenomenon never observed in terrestrial dendritic data. Similar measurements would entail searching for perturbation frequencies at fixed distances from the tip (as first observed by Dougherty and Gollub in terrestrial solution growth of ammonium bromide dendrites). Discussions held this year with scientists at ETH, Zürich, Switzerland, who study noble gas dendrites (xenon, krypton, etc.), proved to them using RIDGE data that noble gas dendrites are much more anisotropic than though previously. Also, recent estimates of surface tension anisotropy, computed by ab initio methods at Sandia-Livermore National Laboratory (J. Hoyt, et al.) show convincingly that hard-sphere fluids (such as the noble gases) have large ca. 3% anisotropies, precisely as verified for PVA dendrites in the IDGE/RIDGE program. Thus the scientific impact of the research conducted under RIDGE continues to assist others in the technical community. It is the purpose of RIDGE to explore topics beyond those considered as the primary science of the IDGE series. Specifically, we have 1) Developed new analysis tools for data extraction from the large number of available USMP-4 video tapes, focusing on transient dendritic growth behavior. 2) Converted selected video sequences from these studies to digital images on CD ROMS for public distribution and use. 3) Continued numerical studies based on the Green’s function method to assess the influence of the side–branch region on the steady–state features. 4) Determined in greater detail the influence of neighboring dendrites through thermal boundary layer interactions. 5) Analyzed IDGE data to determine the behavior of dendrites, including the average side–branch envelope. 6) Developed methods to observe tip dynamics using video methods, especially concerning the possible presence of eigenfrequencies. 7) Established the best procedures for analyzing the shape of PVA dendrites, the profiles for which do not conform to parabolas. 8) Continued the collaborative activities with other groups pursuing modeling efforts associated with dendritic solidification, and assisted

	by access to IDGE data. 9) Analyzed melting sequences from USMP-4 to establish kinetic laws for the melting of prolate spheroids. After completing three scheduled microgravity missions, the IDGE made significant contributions to our understanding of dendritic growth. In the time remaining prior to the termination of the IDGE, we expected to publish several new results that will be of interest to the scientific community that deals with dendritic growth and solidification issues. However, at its scheduled termination date August 31, 1999, we anticipated that a very large backlog of data analysis and reporting would remain uncompleted. These were issues not anticipated a decade earlier in the original IDGE science proposal. Although the IDGE project team accomplished more than was originally expected, or required, a great deal more still remains to be done. The significance of this project is to provide a follow–on to an ongoing MRD flight program, to frame the case that continuing ground–based data reduction and analysis will greatly enhance the understanding of existing or potential flight experiments in material science.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	During the second year of RIDGE, the major area within the IDGE database given major attention: 1) Tip radii and tip speed data were extracted in steady-state PVA dendrites; 2) Side branch spacings in PVA have also been measured from USMP-4 films; 3) Down-linked video data were partially analyzed as displacement-time plots to extract the power spectrum of the moving interface using Lomb periodigrams. RIDGE-supported scientists have shown the presence of dominant eigenfrequencies along the solid-liquid interface near the dendrite tip. These data, augmented by measurements of side-branch coherency, provide support for the trapped-wave hypothesis suggested by J.J. Xu. This is a novel description of the dendritic growth mechanism, which was not supported by experiment until the RIDGE analysis. Sidebranch Envelope To characterize the shape of the sidebranch envelope, the position of the sidebranch tips from the dendrite tip (x coordinate) were linearly regressed in the power law form shown in Equation 1, where the y coordinate is the coordinate normal to the growth axis, or the amplitude of the branches. alpha is the pre-exponential term, and beta is the exponential term. (SEE The annual/final report in .pdf format to view these formula. Added by editor.) . (1) The values of alpha and beta were averaged for each growth and plotted versus supercooling as shown in Figures 3 and 4. It appears that these values are independent of supercooling. Writing these values in the form of Equations 2 and 3: , (2) and . (3) The t-test indicated the convection-free and diffuso-convective environments produced significantly different alpha and beta values (probability of 0.0036 for alpha and 0.00068 for beta). These differences are clear based on the uncertainty measurements alone with out reference to the student t-test. Fig. 3 Pre-exponential term (alpha) for the sidebranch envelope as a function of supercooling for convection-free and diffuso-convective data. alpha appears relatively constant over this range of supercoolings and there is significant statistical difference between the convection and convection-free averages. Fig.4 . Exponential term (beta) for the sidebranch envelope as a function of supercooling for convection-free and diffuso-convective data. beta appears to be relatively constant over this range of supercoolings and there is significant statistical difference between the convection and convection-free averages. The sidebranch envelope could be fitted to a power law equation, where the pre-exponential and exponential terms calculated were found to be significantly different for the convection-free and diffuso-convective results. Convection also appeared to affect the sidebranch envelope in SCN dendrites. These results seem to indicate convection affects the thermal conditions near the solid-liquid interface, which alters the amplitude rather than the spacing of the sidebranches. Task Termination RIDGE was scheduled to end November, 30th, 2003 and it was extended until May 30th, 2004. Both IDGE and RIDGE contributed essentially to the understanding of dendritic solidification with direct implication for castings and ingots. We now have some knowledge of the evolution of the mushy zones, and we would like to explore more on future microgravity experiments to be performed on ISS.
Bibliography Type:	Description: (Last Updated: 04/23/2015)
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Presentation	Glicksman M.E., Lupulescu, A "Mushy Zone Evolution Under Microgravity Conditions" AIAA Aerospace Sciences Meeting, Reno, NV, January 6, 2004 Jan-2004
Presentation	Glicksman M.E. "Dendritic Growth, Charles C. Jones Seminar, Thayer School of Engineering, Dartmouth College" Dartmouth College, NH Jan-2004
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Presentation	Lupulescu, A., Glicksman M.E., Koss, M.B. "Mushy Zone Melting Under Microgravity Conditions" AGH, Faculty and Materials and Ceramics, Krakow, Poland Jul-2004
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